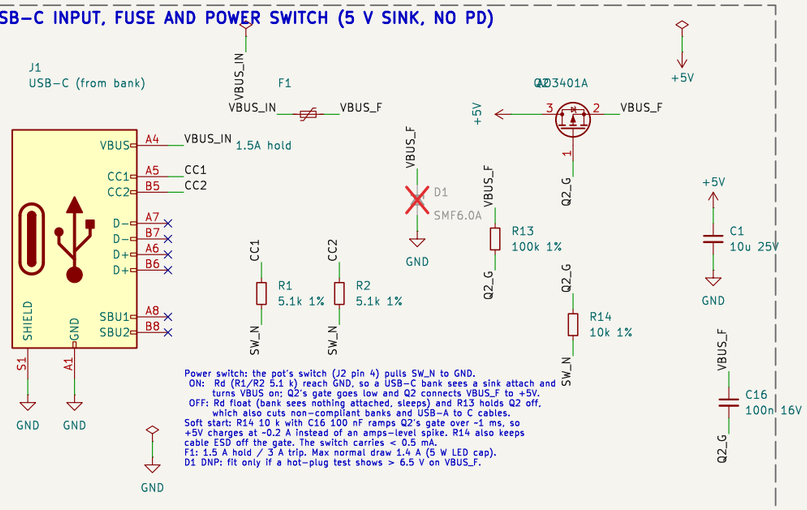
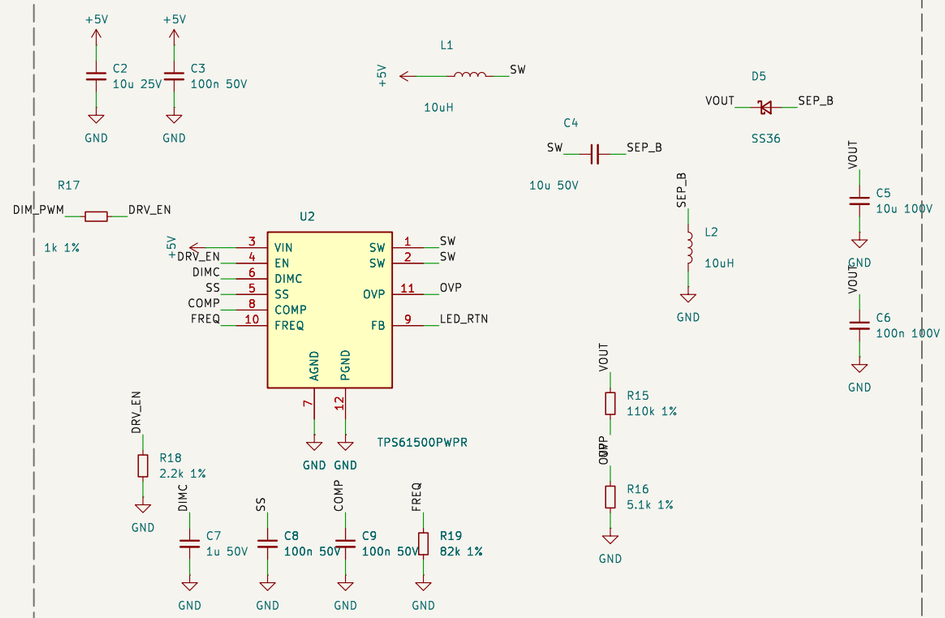
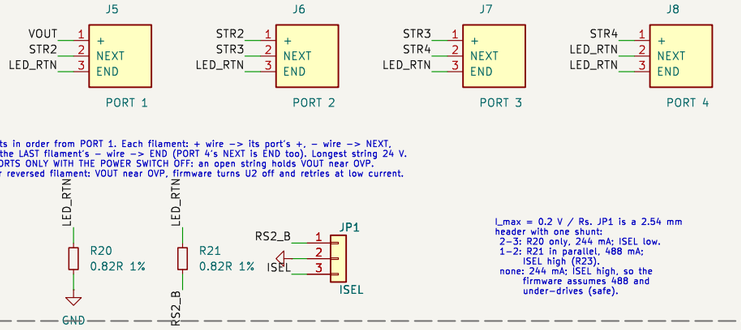
1 USB-C INPUT, FUSE AND POWER SWITCH (5 V SINK, NO PD)



2 LED DRIVER: TPS61500 SEPIC (TI SLVA904), REGULATES 0.2 V ON FB



3 FILAMENT PORTS: ONE SERIES STRING, PUSH-IN TERMINALS



NOTES

Power: 5 V from the bank via Q2. LED power capped at 5 W in firmware (~1.4 A from the bank): 1.7 W lit at prototype brightness; idle ~4 mA with keep-alive, ~0.7 mA without (model: power_model.py).

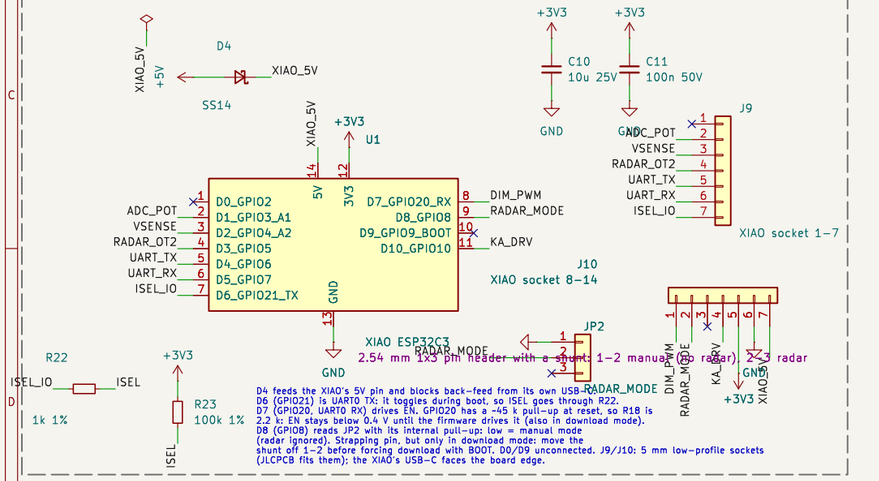
Pins (XIAO):
D1 GPIO3 ADC_POT ADC1_CH3
D2 GPIO4 VSENSE ADC1_CH4, VOUT / 13
D3 GPIO5 RADAR_OT2 wake, high = present
D4 GPIO6 UART_TX -> radar RX (115200 8N1)
D5 GPIO7 UART_RX <- radar OT1/TX
D6 GPIO21 ISEL_IO low = 244 mA (shunt 2-3), via R22
D7 GPIO20 DIM_PWM LEDC 19.5 kHz -> EN (analog dim)
D8 GPIO8 RADAR_MODE JP2 1-2 = manual, 2-3/none = radar
D10 GPIO10 KA_DRV keep-alive gate

Assembly: JLCPCB fits everything incl. through-hole J2-J8 and the XIAO sockets; you plug in the XIAO and the radar. DNP: D1.

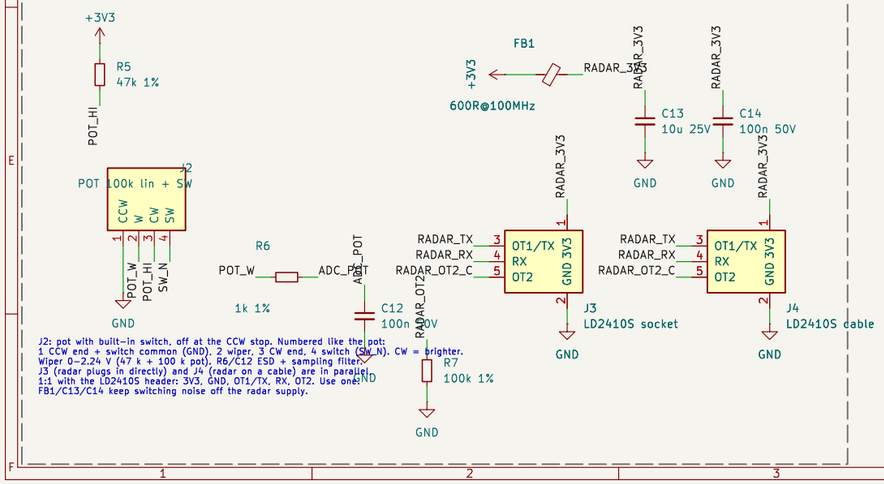
Bring-up (docs/g2_decisions.md): SEPIC at the extremes (one 3 V filament at 244 mA; 24 V string at the 5 W cap). filament temperature in the frame, bank wake via the switch.

SEPIC: +5V -> L1 -> SW (switch) ; C4 couples SW to SEP_B ; L2 SEP_B -> GND ; D5 SEP_B -> VOUT. Works for any string from one 3 V filament (~2.7 V) to 24 V, and C4 blocks DC: no current path from the bank to the LEDs when U2 is off.
f = 1.17 MHz (R19 82 k). OVP = 1.229 x (1 + 110k/5.1k) = 27.7 V (29.1 V worst case); switch sees VIN + VOUT + VD <= 35.1 V worst case (40 V abs max). L1 = L2 = 10 uH, 4.1 A sat. C5 is 100 V (X7S 1210) so it keeps its capacitance and > 2x margin at OVP.
Dimming: 19.5 kHz PWM on EN, filtered inside U2 by DIMC (C7) -> analog FB reference, I_LED = duty x I_max. EN low > 10 ms = shutdown (2 uA). R18 (2.2 k) holds EN below 0.4 V against GPIO20's ~45 k reset pull-up, so U2 stays off through boot and USB flashing.
COMP C9 100 nF. Lowest RHPZ 43 kHz (5 W into a 12 V string, 4.5 V in): keep crossover < 4.3 kHz. SLVA904 measured 1.7 kHz with 15 nF / 4.7 uH; 100 nF / 10 uH is lower still. Tune 15-470 nF on the bench. Checks: sepic_design.py (switch peak 2.2 A vs 3 A min).

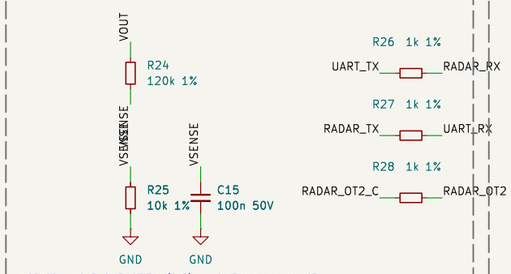
4 MCU: SEEED XIAO ESP32C3 IN 5 mm SOCKETS (bought with pins)



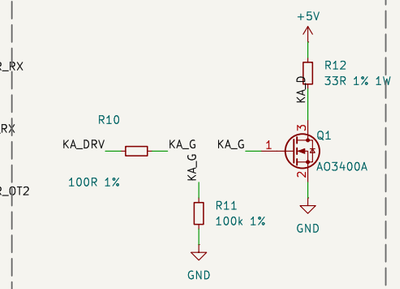
5 POT + POWER SWITCH, RADAR (SOCKET OR PIGTAIL)



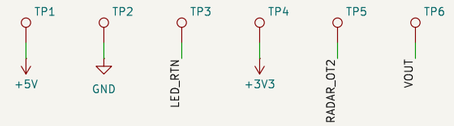
6 STRING VOLTAGE SENSE, RADAR LINE RESISTORS

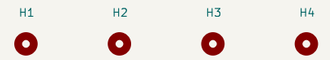


7 BANK KEEP-ALIVE



TP1 +5V TP2 GND TP3 LED_RTN (I = V / 0.82 R) TP4 +3V3 TP5 RADAR_OT2 TP6 VOUT





H1-H4: M3, place to suit the frame

Decisions: docs/g2_decisions.md. Values: hardware/tools/sepic_design.py
XIAO ESP32C3 (socketed), LD2410S radar (socket or pigtail), pot + power switch, keep-alive
Series filament string, TPS61500 SEPIC (TI SLVA904), 244/488 mA, 5 V USB-C input (no PD)

Rob Lim

Sheet: /
File: fgl_g2.kicad_sch

Title: Fused Glass Lighting controller

Size: A3	Date: 2026-10-27	**Rev: G2**
KiCad E.D.A. 9.0.7		Id: 1/1